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Tutorial - 4

1) Permutations

→ Reccurence

$$T(n) = n \cdot T(n-1) + c \qquad T(1) = 1$$

Backwards sub.

$$T(n) = n \cdot T(n-1) + c$$

$$T(n-1) = n \cdot \left[(n-1) \cdot T(n-2) + c \right] + c$$

$$T(n-2) = n(n-1) \cdot \left[(n-2) \cdot T(n-3) + c \right] + cn + c$$

$$\vdots$$

$$T(n) = \frac{n!}{(n-k)!} \cdot T(n-k) + c \sum_{j=0}^{k-1} \frac{n!}{(n-j)!}$$

when $n-k=1$ stop

sub $k = n-1$.

$$T(n) = \frac{n!}{(n-(n-1))!} \cdot T(n-(n-1)) + c \sum_{j=0}^{(n-1)-1} \frac{n!}{(n-j)}$$

$$= n!(1) + c\left(\frac{n!}{n} + \frac{n!}{n(n-1)} + \dots \right)$$

$$\boxed{T(n) = O(n!)}$$

2) Perfect Number

→ $T(n) = T(n-1) + c \qquad T(0) = 1$

Forward

$T(0) = 1$

$T(1) = T(0) + c = 1 + c$

$T(2) = T(1) + c = 1 + 2c$

$T(3) = T(2) + c = 1 + 3c$

$T(n) = 1 + nc$ → Shows same

$T(n) = O(n)$

Backwards

$T(n) = T(n-1) + c$

$T(n-1) = T(n-2) + 2c$

for k

$T(n) = T(n-k) + kc$

$n - k = 0 \quad , \quad k = n$

$T(n) = T(0) + nc$

b) Power Set

$\rightarrow$ $T(n) = T(n-1) + 2.c^{n-1}$, $T(0) = 1$

Backwards

$T(n) = T(n-1) + 2.c^{n-1}$

$T(n-1) = T(n-2) + c.2^{2n-2} + c.2^{n-1}$

$T(n-2) = T(n-3) + c.2^{n-3} + c.2^{n-2} + c.2^{n-1}$

for k

$$T(n) = T(n-k) + c.\sum_{j=0}^{n-1} 2^{n-1-j}$$

$n-k=0$, $n=k$ so

$$T(n) = T(0) + c\sum_{j=0}^{n-1} 2^{n-1-j}$$

$i = n-1-j$

$$T(n) = 1 + c.\sum_{i=0}^{n-1} 2^{i}$$

$$= 1 + c.\frac{2^{n}-1}{2-1}$$

$$= 1 + c.(2^{n}-1)$$

$$\boxed{T(n) = O(2^{n})}$$

General Recursion.

Fibonacci

T(n)

4) Ackerman's

$$A(m,n) = \begin{cases} n+1 & m=0 \\ A(m-1,1) & n=0 \\ A(m-1, A(m,n-1)) & \text{otherwise} \end{cases}$$

m & n non-ve.

→ Base case $m=0$.

$A(0,n) = n+1$

$A(1,n) = n+2$

For $m=1$. & $n=0$.

$A(1,0) = A(0,1) = 2$.

$A(1,1) = A(0, A(1,0)) = A(0,2) = 3$.

$A(2,n) = 2n+3$

For $m=3$

$A(3,0) = 5$

$A(3,1) = 13$

$A(3,2) = 29$

General is $A(3,n) = 2^{n+3} - 3$.

Exponential growth.

We have different complexity for diff. but it's O(exponential).

Fibonacci

$T(n) = T(n-1) + T(n-2) + c$. $T(0) = T(1) = 1$

$\rightarrow$

$T(n) = r^n$

$r^n = r^{n-1} + r^{n-2}$

$r^2 = r + 1 \Rightarrow r^2 - r - 1$

$\downarrow$

quadratic

Use roots

$$\text{roots} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\boxed{\text{roots} = \frac{1 \pm \sqrt{5}}{2}}$$ $\rightarrow$ This is golden ratio.

ϕ.

$r_1 = \phi \qquad r_2 = \hat{\phi}$

$T(A) = A \cdot \phi^n + B \cdot \hat{\phi}^n$

Apply $T(0) =$ & $T(1) = 1$

$$\boxed{A = \frac{\phi}{\sqrt{5}} \qquad B = \frac{-\hat{\phi}}{\sqrt{5}}}$$

$$\boxed{T(n) = O(\phi^n)}$$